

- Hypothetically boosts monoamine actions in the CNS
- May work synergistically with traditional antidepressants

How Long Until It Works (for Depression)

- Can work within days, but therapeutic effects may be delayed for up to 8 weeks

If It Works (for Depression)

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work (for Depression)

- Many patients only have a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)

- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider switching to another antidepressant or adding a different augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- Triiodothyronine is itself an augmenting agent for partial response in depression

Tests

- Periodic assessment of thyroid status
- Administration of triiodothyronine may lead to mild hyperthyroidism with reduced levels of TSH

Side Effects

How Drug Causes Side Effects

- Increases in thyroid hormone concentrations

Notable Side Effects

- Hyperthyroidism (headache, irritability, nervousness, sweating, arrhythmia, increased bowel motility, menstrual irregularities)
- Possible acceleration of bone demineralization, especially in postmenopausal women (controversial)



Life-Threatening or Dangerous Side Effects

- Angina pectoris, congestive heart failure
- Shock

Weight Gain



- Reported but not expected
- May cause weight loss

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Wait
- Wait
- In a few weeks, switch to another agent

Best Augmenting Agents for Side Effects

- Often best to try another treatment prior to resorting to augmentation strategies to treat side effects

Dosing And Use

Usual Dosage Range

- 25–50 $\hat{1}/4$ g/day

Dosage Forms

- Tablet 5 $\hat{1}/4$ g, 25 $\hat{1}/4$ g, 50 $\hat{1}/4$ g

How to Dose

- Initial 25 $\hat{1}/4$ g/day; if no response can increase to 50 $\hat{1}/4$ g/day after 2–4 weeks



Dosing Tips

- Monitor TSH levels to determine efficacy or thyroid actions in the periphery and to guide dosing
- If no effects on depressed mood, may want to discontinue in 8–12 weeks
- To assess efficacy in stabilizing mood in combination with other mood stabilizers, may need to monitor for a few months

Overdose

- Chest pain, increased pulse rate, palpitations, excessive sweating, heat intolerance, nervousness

Long-Term Use

- Has not been evaluated in controlled studies, but long-term treatment of major depressive disorder is generally necessary

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Half-life approximately 2.5 days

Drug Interactions

- Thyroid hormones appear to increase catabolism of vitamin K-dependent clotting factors; patients stabilized on oral anticoagulants who are treated with triiodothyronine should be watched very closely when triiodothyronine is started and may require dose reduction of the oral anticoagulant
- Initiating thyroid replacement therapy may cause increases in insulin or oral hypoglycemic requirements; patients receiving insulin or oral hypoglycemics should be closely watched during initiation of triiodothyronine
- Cholestyramine binds both T4 and T3 in the intestine, thus impairing absorption of these thyroid hormones; therefore, 4 to 5 hours should elapse between administration of cholestyramine and thyroid hormones
- Use of thyroid products with imipramine and other TCAs may increase receptor sensitivity and enhance antidepressant activity; transient cardiac arrhythmias have been observed; thyroid hormone activity may also be enhanced
- Thyroid preparations may potentiate the toxic effects of digitalis
- Use with caution with ketamine; may cause hypertension and tachycardia
- Use with catecholamines may increase their adrenergic effects; careful observation is required

Other Warnings/Precautions

- Use of thyroid hormones in the therapy of obesity, alone or in combination, is not effective at doses within the range of daily hormonal requirements; larger doses may produce serious or even life-threatening manifestations of toxicity, particularly when given in association with sympathomimetic amines
- Use of thyroid hormones is not justified for the treatment of male or female infertility unless accompanied by hypothyroidism
- Use with caution in patients in whom the integrity of the cardiovascular system is suspected, including elderly patients or those with angina pectoris
- Thyroid hormone therapy in patients with concomitant diabetes mellitus or insipidus or adrenal cortical insufficiency aggravates the intensity of their symptoms

Do Not Use

- If patient has uncorrected adrenal cortical insufficiency
- If patient has untreated thyrotoxicosis
- If there is a proven allergy to triiodothyronine

Special Populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- No dose adjustment necessary

Cardiac Impairment

- Use with caution
- Requires dose reduction: initial 5 $\frac{1}{4}$ g; increase by no more than 5 $\frac{1}{4}$ g at 2-week intervals; reduce dose if cardiovascular disease is aggravated

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Thyroid hormone is used safely in infants, children, and adolescents for hypothyroidism
- Not studied for use as adjunct in depression



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Adequate, well-controlled studies in pregnant women have failed to demonstrate risk to the fetus
- Thyroid hormones do not readily cross the placental barrier, and the clinical experience to date does not indicate any adverse effect on fetuses when thyroid hormones are administered to pregnant women

Breast Feeding

- Some drug is found in mother's breast milk
- No known adverse effects but use should be cautious
- Must weigh benefits of breast feeding with risks and benefits of treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art Of Psychopharmacology

Potential Advantages

- Patients with lethargy and fatigue
- Patients with unstable or rapidly fluctuating mood

Potential Disadvantages

- Patients with osteoporosis
- Patients already taking thyroid replacement

Primary Target Symptoms

- Depressed mood

Pearls

- Have periodic monitoring by primary care physician, including neck examination and thyroid palpation
- Generally well tolerated, especially compared to other augmentation options for depression
- May be useful in stabilizing fluctuating mood states as well as improving depressed mood

Suggested Reading

Aronson R , Offman HJ , Joffe RT , et al. Triiodothyronine augmentation in the treatment of refractory depression: a meta-analysis . *Arch Gen Psychiatry* 1996 ;**53** :842 –8 .

Connolly KR , Thase ME. If at first you don't succeed: a review of the evidence for antidepressant augmentation, combination and switching strategies . *Drugs* 2011 ;**71** (1):43 –64 .

Garlow SJ , Dunlop BW , Ninan PT , Nemeroff CB. The combination of triiodothyronine (T3) and sertraline is not superior to sertraline monotherapy in the treatment of major depressive disorder . *J Psychiatr Res* 2012 ;**46** (11):1406 –13 .

.....
Hage MP , Azar ST. The link between thyroid function and depression . *J
Thyroid Res* 2012 ;**2012** :590648 .
.....

Nierenberg AA , Fava M , Trivedi MH , et al. A comparison of lithium and
T(3) augmentation following two failed medication treatments for
depression: a STAR*D report . *Am J Psychiatry* 2006 ;**163** :1519 –30 .

Trimipramine

Therapeutics

Brands

- Surmontil
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Tricyclic antidepressant (TCA)
- Serotonin and norepinephrine/noradrenaline reuptake inhibitor

Commonly Prescribed for

(bold for FDA approved)

- **Depression**
- **Endogenous depression**
- Anxiety
- Insomnia

- Neuropathic pain/chronic pain
- Treatment-resistant depression

How the Drug Works

- Boosts neurotransmitters serotonin and norepinephrine/noradrenaline
- Blocks serotonin reuptake pump (serotonin transporter), presumably increasing serotonergic neurotransmission
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Presumably desensitizes both serotonin 1A receptors and beta adrenergic receptors
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, trimipramine can increase dopamine neurotransmission in this part of the brain

How Long Until It Works

- May have immediate effects in treating insomnia, agitation, or anxiety
- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all

- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment of depression is complete remission of current symptoms as well as prevention of future relapses
- The goal of treatment of chronic neuropathic pain is to reduce symptoms as much as possible, especially in combination with other treatments
- Treatment of depression most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Treatment of chronic neuropathic pain may reduce symptoms, but rarely eliminates them completely, and is not a cure since symptoms can recur after medicine is stopped
- Continue treatment of depression until all symptoms are gone (remission)
- Once symptoms of depression are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders and chronic pain may also need to be indefinite, but long-term treatment is not well studied in these conditions

If It Doesn't Work

- Many depressed patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other depressed patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- Lithium, buspirone, thyroid hormone (for depression)
- Gabapentin, tiagabine, other anticonvulsants, even opiates if done by experts while monitoring carefully in difficult cases (for chronic pain)

Tests

- Baseline ECG is recommended for patients over age 50

* Since tricyclic and tetracyclic antidepressants are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- ECGs may be useful for selected patients (e.g., those with personal or family history of QTc prolongation; cardiac arrhythmia; recent myocardial infarction; uncompensated heart failure; or taking agents that prolong QTc interval such as pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin, etc.)
- Patients at risk for electrolyte disturbances (e.g., patients on diuretic therapy) should have baseline and periodic serum potassium and magnesium measurements

Side Effects

How Drug Causes Side Effects

- Anticholinergic activity may explain sedative effects, dry mouth, constipation, and blurred vision
- Sedative effects and weight gain may be due to antihistamine properties
- Blockade of alpha adrenergic 1 receptors may explain dizziness, sedation, and hypotension
- Cardiac arrhythmias and seizures, especially in overdose, may be caused by blockade of ion channels

Notable Side Effects

- Blurred vision, constipation, urinary retention, increased appetite, dry mouth, nausea, diarrhea, heartburn, unusual taste in mouth, weight gain
- Fatigue, weakness, dizziness, sedation, headache, anxiety, nervousness, restlessness
- Sexual dysfunction (impotence, change in libido)
- Sweating, rash, itching



Life-Threatening or Dangerous Side Effects

- Paralytic ileus, hyperthermia (TCAs + anticholinergic agents)
- Lowered seizure threshold and rare seizures

- Orthostatic hypotension, sudden death, arrhythmias, tachycardia
- QTc prolongation
- Hepatic failure, drug-induced parkinsonism
- Increased intraocular pressure
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Many experience and/or can be significant in amount
- Can increase appetite and carbohydrate craving

Sedation



- Many experience and/or can be significant in amount
- Tolerance to sedative effects may develop with long-term use

What to Do About Side Effects

- Wait
- Wait
- Wait

- Lower the dose
- Switch to an SSRI or newer antidepressant

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 50–150 mg/day

Dosage Forms

- Capsule 25 mg, 50 mg, 100 mg

How to Dose

- Initial 25 mg/day at bedtime; increase by 75 mg every 3–7 days
- 75 mg/day in divided doses; increase to 150 mg/day; maximum 200 mg/day; hospitalized patients may receive doses up to 300 mg/day



Dosing Tips

- If given in a single dose, should generally be administered at bedtime because of its sedative properties
- If given in split doses, largest dose should generally be given at bedtime because of its sedative properties

- If patients experience nightmares, split dose and do not give large dose at bedtime
- Patients treated for chronic pain may only require lower doses
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder, and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- Death may occur; CNS depression, convulsions, cardiac dysrhythmias, severe hypotension, ECG changes, coma

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects
- Even with gradual dose reduction some withdrawal symptoms may appear within the first 2 weeks
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation

- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

Pharmacokinetics

- Substrate for CYP450 2D6, 2C19, and 2C9
- Half-life approximately 7–23 hours



Drug Interactions

- Tramadol increases the risk of seizures in patients taking TCAs
- Use of TCAs with anticholinergic drugs may result in paralytic ileus or hyperthermia
- Fluoxetine, paroxetine, bupropion, duloxetine, and other CYP450 2D6 inhibitors may increase TCA concentrations
- Cimetidine may increase plasma concentrations of TCAs and cause anticholinergic symptoms
- Phenothiazines or haloperidol may raise TCA blood concentrations
- May alter effects of antihypertensive drugs; may inhibit hypotensive effects of clonidine
- Use with sympathomimetic agents may increase sympathetic activity
- Methylphenidate may inhibit metabolism of TCAs
- Activation and agitation, especially following switching or adding antidepressants, may represent the induction of a bipolar state,

especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of trimipramine

Other Warnings/Precautions

- Add or initiate other antidepressants with caution for up to 2 weeks after discontinuing trimipramine
- Generally, do not use with MAOIs, including 14 days after MAOIs are stopped; do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing trimipramine, but see Pearls
- Use with caution in patients with history of seizures, urinary retention, angle-closure glaucoma, hyperthyroidism
- TCAs can increase QTc interval, especially at toxic doses, which can be attained not only by overdose but also by combining with drugs that inhibit TCA metabolism via CYP450 2D6, potentially causing torsade de pointes-type arrhythmia or sudden death
- Because TCAs can prolong QTc interval, use with caution in patients who have bradycardia or who are taking drugs that can induce bradycardia (e.g., beta blockers, calcium channel blockers, clonidine, digitalis)
- Because TCAs can prolong QTc interval, use with caution in patients who have hypokalemia and/or hypomagnesemia or who are taking drugs that can induce hypokalemia and/or magnesemia (e.g.,

diuretics, stimulant laxatives, intravenous amphotericin B, glucocorticoids, tetracosactide)

- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is recovering from myocardial infarction
- If patient is taking agents capable of significantly prolonging QTc interval (e.g., pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin)
- If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- If patient is taking drugs that inhibit TCA metabolism, including CYP450 2D6 inhibitors, except by an expert

- If there is reduced CYP450 2D6 function, such as patients who are poor 2D6 metabolizers, except by an expert and at low doses
- If there is a proven allergy to trimipramine

Special Populations

Renal Impairment

- Use with caution; may need to lower dose

Hepatic Impairment

- Use with caution; may need to lower dose

Cardiac Impairment

- Baseline ECG is recommended
- TCAs have been reported to cause arrhythmias, prolongation of conduction time, orthostatic hypotension, sinus tachycardia, and heart failure, especially in the diseased heart
- Myocardial infarction and stroke have been reported with TCAs
- TCAs produce QTc prolongation, which may be enhanced by the existence of bradycardia, hypokalemia, congenital or acquired long QTc interval, which should be evaluated prior to administering trimipramine
- Use with caution if treating concomitantly with a medication likely to produce prolonged bradycardia, hypokalemia, slowing of

intracardiac conduction, or prolongation of the QTc interval

- Avoid TCAs in patients with a known history of QTc prolongation, recent acute myocardial infarction, and uncompensated heart failure
- TCAs may cause a sustained increase in heart rate in patients with ischemic heart disease and may worsen (decrease) heart rate variability, an independent risk of mortality in cardiac populations
- Since SSRIs may improve (increase) heart rate variability in patients following a myocardial infarct and may improve survival as well as mood in patients with acute angina or following a myocardial infarction, these are more appropriate agents for cardiac population than tricyclic/tetracyclic antidepressants

* Risk/benefit ratio may not justify use of TCAs in cardiac impairment

Elderly

- Baseline ECG is recommended for patients over age 50
- May be more sensitive to anticholinergic, cardiovascular, hypotensive, and sedative effects
- Initial dose 50 mg/day; increase gradually up to 100 mg/day
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents